

MCELROY AFLD, CO, USA

WMO: 720262

Lat: 40.054N

Lon: 106.369W

Elev: 7411

StdP: 11.16

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94076

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-11.2	-25.7	1.7	-16.0	-17.8	2.7	-7.8	25.0	34.4	21.7	35.8	0.2	90	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.8	88.5	53.3	85.5	53.1	81.6	52.1	58.5	74.2	57.1	73.5	55.7	73.0	9.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	82.9	63.1	52.2	76.6	61.4	50.1	70.8	59.9	29.1	75.5	27.9	73.8	26.8	71.9	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	18.8	16.3		-26.9	93.2	3.1	3.3	-29.2	95.6	-31.0	97.6	-32.7	99.4	-34.9	101.9
			DB	-26.0	60.8	5.8	2.0	-30.2	62.3	-33.6	63.4	-36.9	64.6	-41.2	66.0
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	13.5	21.3	32.3	41.0	48.8	58.6	65.1	61.8	54.7	42.2	31.3	17.0
	DBStd	19.05	12.57	12.54	9.60	6.79	6.50	5.66	4.32	3.96	5.38	7.76	9.38	12.07
	HDD50	4720	1131	804	548	274	100	7	0	0	16	253	562	1025
	HDD65	8938	1596	1223	1012	719	501	202	53	112	310	708	1012	1490
	CDD50	1339	0	0	1	5	65	264	468	367	159	10	0	0
	CDD65	82	0	0	0	0	0	9	56	14	3	0	0	0
	CDH74	3709	0	0	0	3	85	926	1538	858	297	2	0	0
	CDH80	1089	0	0	0	0	7	287	555	196	44	0	0	0
Wind	WSAvg	4.1	2.9	4.3	4.7	5.9	5.2	5.0	3.7	3.6	3.7	4.0	3.7	3.0
Precipitation	PrecAvg	13.5	0.9	0.9	0.9	1.2	1.3	1.1	1.3	1.5	1.7	1.0	1.0	0.6
	PrecMax	18.1	2.3	1.8	1.5	2.1	3.7	2.6	2.4	2.9	3.2	1.9	1.5	1.2
	PrecMin	9.0	0.3	0.3	0.5	0.6	0.3	0.2	0.4	0.5	0.7	0.2	0.6	0.2
	PrecStd	2.4	0.5	0.5	0.3	0.4	0.7	0.6	0.6	0.6	0.7	0.5	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	54.1	67.8	72.7	81.0	91.0	94.6	88.4	85.6	73.0	63.2	51.7
		MCWB	33.4	40.2	43.3	44.5	48.9	51.8	55.6	54.1	53.8	46.4	41.6	36.5
	2%	DB	38.5	46.4	60.7	68.0	76.7	87.8	90.2	85.5	80.8	70.1	59.2	43.2
		MCWB	31.6	35.4	40.0	43.1	47.9	51.4	54.5	53.6	50.3	45.5	39.8	32.3
	5%	DB	36.2	43.3	54.4	63.2	72.6	82.1	86.4	81.6	77.4	64.4	54.3	37.5
		MCWB	30.9	33.5	38.0	41.0	46.5	50.5	54.1	52.5	48.7	43.7	38.4	30.8
	10%	DB	33.8	39.1	49.8	59.0	69.5	79.4	81.9	79.3	73.3	60.5	49.5	35.6
		MCWB	29.1	31.7	36.5	39.8	45.4	49.8	53.6	52.3	48.2	42.1	36.7	30.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	41.5	44.0	46.3	51.7	56.9	61.8	59.8	56.3	50.4	43.4	37.2
		MCDB	40.8	52.0	63.3	66.4	73.9	72.1	77.0	72.8	73.5	63.2	60.1	48.6
	2%	WB	33.2	37.1	41.5	44.1	49.7	54.9	59.5	58.0	54.3	47.5	40.9	33.8
		MCDB	37.1	45.3	57.8	63.2	70.6	75.0	75.7	72.3	70.5	63.9	56.3	40.9
	5%	WB	31.0	34.8	39.1	42.3	48.1	53.4	57.9	56.5	52.6	45.3	38.9	31.8
		MCDB	35.5	42.4	52.6	59.7	67.8	75.0	74.8	71.7	68.1	61.7	52.6	38.1
	10%	WB	28.6	32.4	36.8	40.8	46.4	51.9	56.6	55.1	50.6	43.2	36.7	29.9
		MCDB	33.1	39.0	48.1	56.7	64.5	74.5	74.1	71.0	67.3	58.9	47.9	34.4
Mean Daily Temperature Range	5% DB	MDBR	24.7	24.4	25.6	27.9	30.8	38.3	36.8	36.1	37.2	31.0	28.1	23.7
		MCDBR	23.1	24.1	34.2	39.7	41.0	45.1	43.3	40.8	43.8	41.5	36.7	24.0
		MCWBR	17.4	15.0	16.5	18.0	15.9	15.7	13.7	14.5	17.0	20.1	20.2	16.3
	5% WB	MCDBR	21.4	21.6	31.5	34.1	36.8	38.4	33.5	31.2	34.3	36.0	32.9	22.0
		MCWBR	17.0	13.7	15.9	15.9	15.2	14.8	11.9	11.9	13.9	17.9	18.5	15.5
Clear-Sky Solar Irradiance	taub	0.224	0.236	0.260	0.298	0.310	0.315	0.344	0.340	0.292	0.266	0.241	0.229	
	taud	2.431	2.375	2.403	2.417	2.490	2.480	2.445	2.438	2.588	2.625	2.579	2.473	
	Ebn at Noon	309	318	319	310	306	303	293	292	304	304	301	298	
	Edh at Noon	24	30	33	36	34	34	35	35	28	24	21	22	
All-Sky Solar Radiation	RadAvg	785	1081	1529	1787	1943	2321	2112	1825	1611	1161	810	646	
	RadStd	40	67	119	132	148	143	102	92	109	98	76	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page